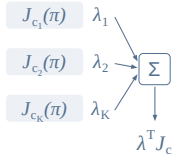


Penalty view

$$\max_{\pi} J_r(\pi) - \lambda^T J_c(\pi)$$

Weighted costs



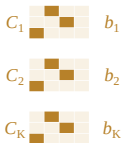
Exact settings $\cdot c = 0$

fixed $\beta \rightarrow$ fixed penalty
adaptive $\beta \rightarrow$ Lagrangian

SafeOT-Dual

One shared price construction

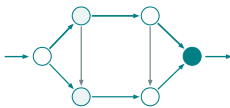
Joint budgets



Hard support H



Shared graph

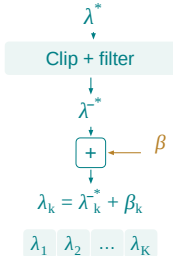


$$\min_F - \langle A, F \rangle + \varepsilon KL(F \parallel \hat{F})$$

flow balance

$$\langle C_k, F \rangle \leq b_k$$

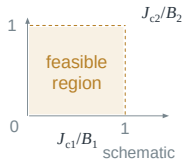
Safety price



Constraint view

$$\max_{\pi} J_r(\pi) \text{ s.t. } J_c(\pi) \leq B$$

Joint budgets



Graph-price limits

under assumptions
 $\varepsilon \rightarrow 0$ LP-derived price
 $\varepsilon \rightarrow \infty$ clipped active-set

One policy-update interface

$$\tilde{A} = (A_r - \sum_k \lambda_k A_{c_k}) / (1 + \sum_k \lambda_k)$$

$\rightarrow$ PPO / TRPO

Graph space

